Quiz 6

Q1. Breakdown mechanism:

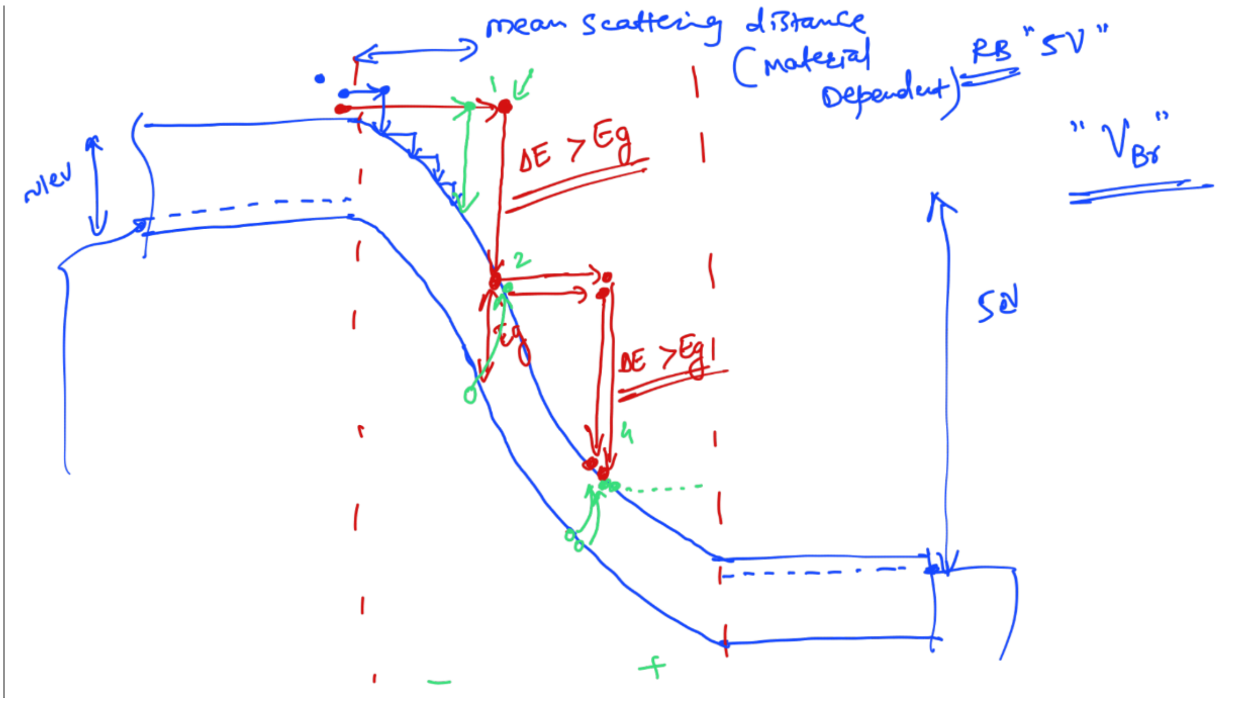
a) Describe avalanche breakdown in brief using band diagram and relating KE with bandgap. **(2 Marks)**

Answer: In reverse bias, minority carriers (electron from p-side and holes from n-side) gain kinetic energy from the electric field inside the depletion width while moving from p-side to n-side (for electrons) and n-side to p-side (for holes) as shown in figure.

These carriers move in the physical space (within Si covalent bonds) and get scattered by colliding with atoms. When the carriers scatter, they lose the KE they gained. If carriers are scattered over a small distance, the KE released is not enough (<Eg) to break the bonds and create electron-hole pair (EHP). The KE released goes into lattice vibration (heat).

If the carriers scattering distance is large, the carrier gain large KE.

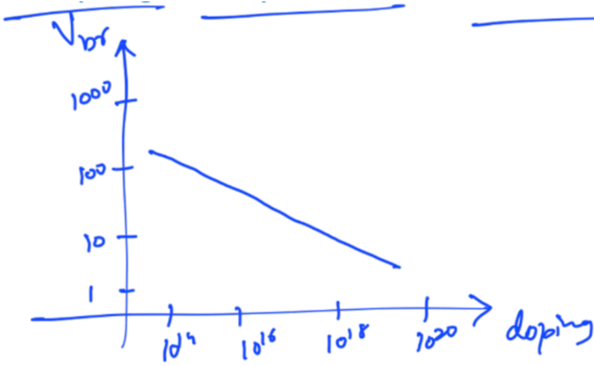
When the carrier's kinetic energy exceeds the bandgap energy ($E_k > E_g$), it can break the co-valent bond and creates EHP i.e. electron goes from valence band to the conduction band. This is knows as **impact ionization (II)**. Same process repeats until carriers cross the depletion width. In the process, due to the II, carriers get multiplied i.e. 1 -> 2->4-> 8 …. leading to large carriers generation leading to large current in RB i.e known avalanche breakdown.



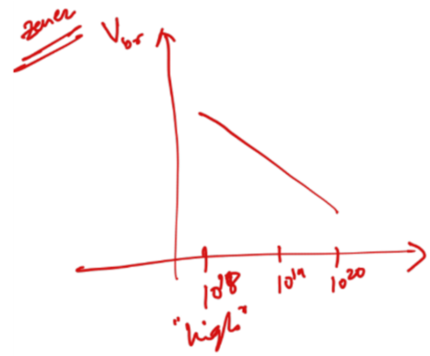
b) If the material doping is increased, how does it affect the Avalanche and Zener Breakdown voltage change? **(2 Marks)**

Answer:

Breakdown Type	Effect on Breakdown Voltage when Doping ↑
Avalanche	**Decreases** – Higher doping → higher electric field for same voltage → breakdown occurs at lower voltage
Zener	**Decreases** – Higher doping → narrower depletion width → easier tunneling at lower voltage



Avalanche Breakdown



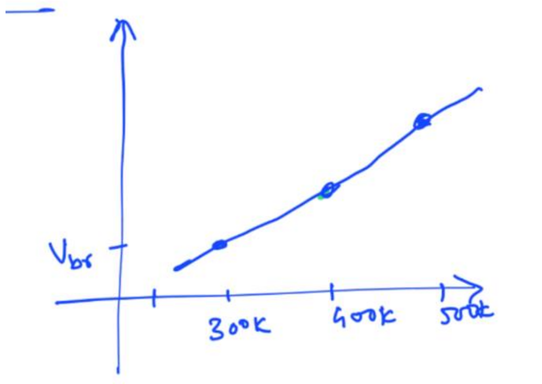
Zener Breakdown

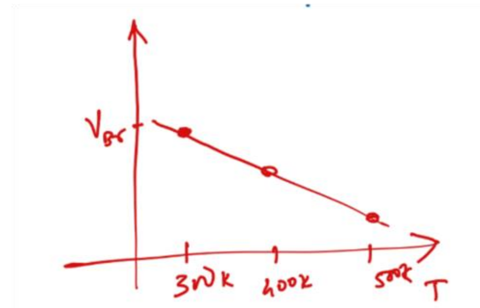
c) If a diode is given to you randomly, can you go into the lab and find out if it is Zener diode or Avalanche diode? **(2 Marks)**

Yes, we can conduct some tests to determine it.

1) Temperature dependence; Avalanche and Zener breakdown have opposite relationship with temperature, we use this information to find out the type of diode. As the temperature of a semiconductor increases, its energy band gap (Eg) narrows. As temperature increases, the depletion region narrows, making Zener breakdown more likely to occur at lower voltage. In other words, Zener voltage decreases as temperature increases. (Negative temperature coefficient)

Increase in temperature increases the crystal lattice vibration. Therefore, an increase in temperature makes it more difficult for electrons and holes to flow. Their mean scattering distance decreases which leads to lower KE being released. Avalanche breakdown generates electron-hole pairs when electrons accelerated by an electric field collide with atoms in a crystal. An increase in temperature causes avalanche breakdown to occur at higher voltage. In other words, avalanche breakdown voltage increases as temperature increases. (Positive temperature coefficient)





Avalanche Breakdown

Zener Breakdown

2.If the breakdown occurs at a higher reverse voltage and involves a more gradual increase in current, it may indicate avalanche breakdown [If the measured breakdown voltage is low (e.g., < 5V), the dominant mechanism is the Zener effect.].

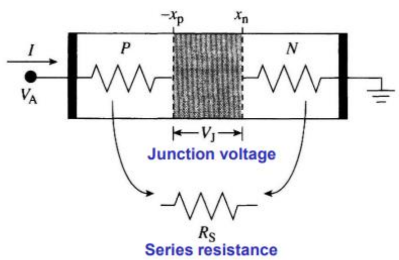
Q2. Explain how the quasi-neutral region (i.e., series resistance) affects the diode IV when a large FB voltage is applied V > Vbi? **(4 Marks)**

Answer: At high forward bias ($V > V_{bi}$), the applied voltage divides between the junction and the series resistance R_s of the quasi-neutral regions (bulk semiconductor and contacts):

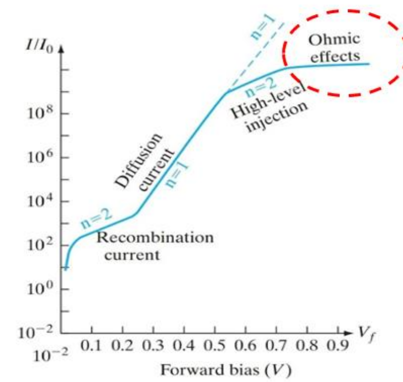
$$V = V_j + IR_s,$$

where V_j is the actual voltage across the junction.

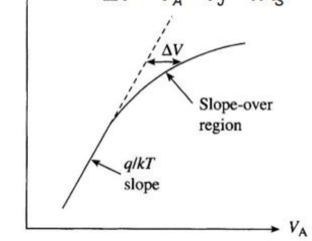
- At low currents, IR_s is negligible → ideal exponential behavior $I = I_s e^{V_j/V_T}$.
- As current increases, IR_s becomes significant → junction voltage $V_j = V - IR_s$ is lower than the applied V.



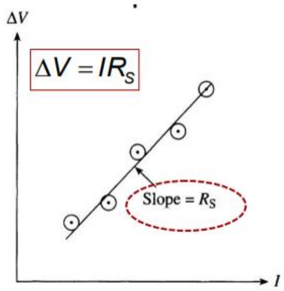
At current levels where IR_s becomes comparable to V_A, the applied voltage drop is reduced to $\boxed{V_J = V_A - IR_S}$



$$\Delta V = V_A - V_J = IR_S$$



$$\boxed{\Delta V = IR_S}$$



$$\boxed{I = I_0(e^{qV_A/kT} - 1)}$$

$$V_A \rightarrow V_{bi}$$

$$I = I_0 e^{qV_J/kT}$$
$$= I_0 e^{q(V_A - IR_S)/kT}$$